

Worksheet No. - 1

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Branch: MCA general

Semester: 3rd

Subject Name: Backend Technology.

Subject Code: CONT_25CAP-702

UID: 25MCA20023

Section: Group:1/A

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Experiment 1: Create employee salary program using node.js.

1.Aim:

To develop a Node.js program using the `readline` module that accepts employee details from the user, calculates the salary components (HRA, DA, Transport Allowance, and Gross Pay), and displays the salary report on the console.

2.Objective:

1. To understand the use of the `readline` module in Node.js for accepting user input.
2. To collect employee details such as Employee ID, Name, Department, and Basic Pay from the user.
3. To perform salary calculations using JavaScript arithmetic operations.
4. To calculate HRA, DA, Transport Allowance, and Gross Pay based on the employee's basic pay.
5. To display the employee salary details in a well-formatted report on the console.
6. To gain practical experience in developing console-based Node.js applications.

3.Task to be done :

- Create a Node.js program using the `readline` module.
- Accept employee details from the user:
 - Basic Pay
 - Convert the entered Basic Pay into a numeric value.
 - Calculate the following salary components:
 - HRA = 20% of Basic Pay
 - DA = 10% of Basic Pay
 - Transport Allowance = 3000 (fixed)
 - Gross Pay = Basic Pay + HRA + DA + Transport Allowance
- Display the employee details and calculated salary components in a well-formatted report on the console.
- Close the `readline` interface after displaying the output.

4.Online Resources:

- Node.js Official Documentation
- EJS Official Documentation
- MDN Web Docs (HTML, CSS, JavaScript)
- W3Schools – HTML, CSS, JavaScript
- GeeksforGeeks – Node.js and Express.js Tutorials

Software Requirements

- Node.js (Latest LTS Version)
- Visual Studio Code
- Google Chrome / Microsoft Edge / Mozilla Firefox
- Command Prompt / Windows PowerShell / Terminal
- npm (Node Package Manager)

Standard Tools Used

- Node.js
- HTML5
- CSS3
- JavaScript
- File System (fs) Module
- Browser (Chrome/Edge/Firefox) for user input and displaying the output

5.Procedure/ Algorithm:

Part1:Display on Terminal

```
const readline = require("readline");

const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

rl.question("Enter Employee ID: ", function(id) {

  rl.question("Enter Employee Name: ", function(name) {

    rl.question("Enter Department: ", function(department) {

      rl.question("Enter Basic Pay: ", function(basicPay) {

        basicPay = Number(basicPay);

        let hra = basicPay * 0.20;
```

```
let da = basicPay * 0.10;
let transport = 3000;           // Fixed Transport Allowance
let grossPay = basicPay + hra + da + transport;.....
```

Output:

```
PS C:\Users\ASUS\EmployeeSalary> node server.js
Enter Employee ID: 101
Enter Employee Name: Anjali kumari
Enter Department: mca
Enter Basic Pay: 65000

===== EMPLOYEE SALARY DETAILS =====
Employee ID       : 101
Employee Name     : Anjali kumari
Department       : mca
Basic Pay        : 65000
HRA (20%)       : 13000
DA (10%)        : 6500
Transportation Allow. : 3000
-----
Gross Pay        : 87500
=====
PS C:\Users\ASUS\EmployeeSalary> |
```

Part 2: Display on Browser

```
<!-- Charts Section -->
```

```
<div class="row mt-4">
```

```
<!-- Bar Chart -->
```

```
<div class="col-md-6">
```

```
<div class="salary-card">
```

```
<h3 class="text-primary">
```

```
<i class="bi bi-bar-chart-fill"></i>
```

Salary Summary

```
</h3>
```

```
<div style="height:250px;">
```

```
<canvas id="salaryChart"></canvas>
```

```
</div>
```

```
</div>
```



</div>

<!-- Pie Chart -->

<div class="col-md-6">

<div class="salary-card">

<h3 class="text-success">

<i class="bi bi-pie-chart-fill"></i>

Salary Distribution

</h3>

<div style="height:250px;">

<canvas id="pieChart"></canvas>

</div>

</div>

</div>

</div>.....

6.Output:

ABC Technologies Pvt Ltd

Employee Salary Management System

MCA Node.js Mini Project



Anjali kuamri

student

ID: 1001

Department: mca

9991088359

anjali@com

Employee Information

Employee ID	1001
Employee Name	Anjali kuamri
Age	24
Gender	female
Date of Birth	21/07/20013



Contact Information

Email	anjali@com
Phone	9991088359
Address	kitadih
City	jamshedpur

Job Information

Department	mca
Designation	student
Joining Date	27 june
Employee Type	full-time



Monthly Salary
₹ undefined



Annual Salary
₹ undefined



Experience
undefined



Current Time
7/27/2026, 2:08:58 PM

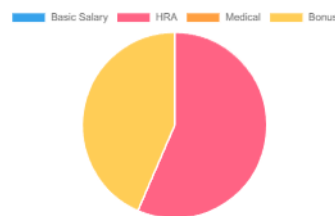
₹ Salary Details

Basic Salary	₹ undefined
HRA	₹ 13000
Medical Allowance	₹ undefined
Bonus	₹ 10000
Total Salary	₹ undefined

Salary Summary



Salary Distribution



Print Report

6.Learning outcomes (what I have learnt) :

- Learned how to build a basic employee salary system using Node.js.
- Understood file handling and storing employee data.
- Learned how to calculate salary details like HRA, DA, and net salary.
- Created a dynamic dashboard to display employee information.
- Used Bootstrap and Chart.js to design a responsive interface and graphs.
- Improved programming, debugging, and problem-solving skills.